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Feasibility of laser-induced breakdown spectroscopy as a direct raw bile analysis tool for screening of gallbladder cancer

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Laser-induced breakdown spectroscopy (LIBS) was explored for the first time for discrimination of gallbladder (GB) polyp and GB cancer (GBC) by direct measurement of raw bile juice without pretreatment. Initially, for reproducible LIBS measurement, a laser-patterned silicon wafer (LPSW) incorporating 40 × 40 crossed trenches in an area of 1 × 1 cm^2 was used to spread bile juice as homogeneous as possible over the surface without substantial localization, and LiCl was employed as an internal standard. Mg, Ca, Na, and K peaks were dominant in the spectra of raw bile samples, with lower peak areas in cases of GBC. The lower elemental concentrations in GBC samples probably result from dilution of bile by a degraded water absorption ability of the thickened GB wall due to cancer and carcinogenesis. In particular, the Mg and K peak intensities were further lower and improved discrimination of GBC against normal/GB polyp. Moreover, discrimination was distinct when using the ratio of Na and K peak areas. Misbalancing of Na and K in cell metabolism induced by cancer progression was presumed to be the origin of enhanced discrimination. Overall, the LIBS-based direct bile analysis scheme is a simple and potential tool for screening of GBC.

Introduction

Infrared (IR) spectroscopy has been previously studied for identification of gallbladder cancer (GBC) against gallbladder (GB) polyp[1] through the measurement of human bile, a valuable specimen potentially indicating pathological conditions of the GB. For analysis, each raw bile sample was separated into three fractions of aqueous, organic, and amphiphilic aggregate phases using a chloroform/methanol extraction solution, and IR spectra of each phase were separately acquired. The aqueous, organic, and amphiphilic phases contained mostly bile acid/bile salts, cholesterol/lipids, and proteins, respectively. Three-phase separation was performed to acquire more component-discernible IR features by decreasing composition complexity and to utilize the characteristic IR information of each separated phase to enhance GBC identification. Nonetheless, the demonstrated IR spectroscopic method would be less attractive for practical clinical analysis since it requires time- and reagent-consuming bile separation. An analytical tool able to measure raw bile samples directly and rapidly would be desirable.

In this context, laser-induced breakdown spectroscopy (LIBS) is a candidate, since elements are extensively involved in diverse biological functions and metabolic pathways, and their compositions in medical specimens vary depending on related disease conditions. Also, the ability of LIBS to allow fast and direct sample analysis with no or minimized pretreatment would be of great appeal in clinical practice. Therefore, LIBS has been studied for identification of various diseases including cancer and melanoma as summarized in Table 1, and diverse specimens such as tissues, serums, and finger nails were analyzed.[2–11] The elemental information of Mg, Ca, Na, and/or K has been more commonly adopted in the studies of the listed reports. As seen, LIBS is a potential tool for biomedical diagnosis.

This study explored LIBS as a potential disease diagnosis tool in a hepato-biliary-pancreatic area and demonstrated the LIBS-based bile analysis scheme as a new method for identification of GBC. This research was driven by the expectation that compositions of bile juice samples would vary depending on GB disease. Imaging-based methods such as sonography[12–14] and computerized tomography (CT)[14] are common choices for cancer identification; however, the identification based on physical appearance in the obtained images would not be accurate enough, since it is predominantly dependent on just the size of polypoid lesions. Further, examination on a biopsy of epithermal cell tissue typically provides the final medical

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Table 1 Summary of research that utilized elemental analysis to study diverse human and animal diseases

Subject	Disease	Employed sample	Studied elements	Ref.
Human	Cancer	Breast, colon, tongue, and larynx tissues	Ca, K, Mg, and Na	2
	Cancer	Cervix tissue	Al, Ca, K, Mg, and Na	3
	Cancer	Colon tissue	Ce, Cr, Hg, and Pb	4
	Wilson's disease	Liver tissue	C and Cu	5
	Thyroidism	Fingernail	Ca, K, and Na	6
	Lymphoma, myeloma	Human serum	Ca, H, K, N, and Na	7
Animal	Cutaneous melanoma	Mouse skin tissue	Ca and Mg	8
	Cutaneous melanoma	Mouse skin tissue	Ca and Mg	9
	Melanoma	Mouse serum, lung, lymph node, and spleen	Ca, Mg, and Na	10
	Liver fibrosis	Rat liver tissue	C, Ca, and Mg	11

opinion.[15–17] In addition to the conventional diagnosis, a rapid, non-invasive, and supportive method enabling improvement of the accuracy as well as simplicity of diagnosis by chemical analysis of body fluids would be valuable. In this sense, the direct LIBS analysis of bile meets the demand.

For exploratory investigation, 30 raw bile juice samples obtained from 6 normal, 17 GB polyp, and 7 GBC subjects were employed in this study. The spectral features in the LIBS spectra of all bile juice samples were initially characterized, and the areas of major element peaks in normal, GB polyp, and GBC samples were compared to find out whether the peak areas were different according to the diseases or not. Also, a peak intensity ratio between two elements would be a feasible indicator for discrimination of GB disease since multiple elements simultaneously play important roles in various metabolic pathways, and their balance often influences biological functions. Therefore, two different element peaks were chosen out of the major element peaks in a combinatorial fashion, and subsequent peak area ratios were investigated in relation with GB diseases. Finally, origins of the dissimilar elemental peak areas and their ratios in GB polyps and GBC were discussed.

Experimental

Collection of bile juice samples

With consent from each subject, bile juice samples were collected after cholecystectomy at Hanyang University Hospital (Seoul, South Korea). The acquired bile samples were immediately frozen and stored below −20 °C to prevent further denaturation. A total of 30 bile samples were obtained from 7 GBC, 17 GB polyp, and 6 healthy subjects. The healthy subjects were liver donors for transplantation. The colors of bile samples varied from yellowish brown to deep green. The purpose and content of our research were clearly explained to each patient, and the overall study was performed under the guidelines of the Institutional Review Board of Hanyang University (HYI-18-227-3).

Preparation of the LPSW, and LIBS measurement of raw bile samples using the LPSW

The LPSW demonstrated in a former publication[18] was prepared as a substrate for LIBS measurement of raw bile samples. A lattice pattern was engraved over a 1 × 1 cm^2 area on a Si wafer using a diode-pumped solid-state laser (DPSSL, 1064 nm, 7 ns pulse duration, 270 μJ pulse energy, DTL-329QT, Laser-export Co. Ltd) and a motorized *xyz* stage (L-406.40DD10, L-406.40DD10, and L-306; Physik Instrumente GmbH & Co. KG). The DPSSL was operated at a repetition rate of 1 kHz, and the stage was moved horizontally at 1 mm s^{-1}. 90° crossing 40 × 40 trenches were produced to form the lattice pattern shown in Fig. 1(a). As shown in the magnified image (Fig. 1(b)), the space between two nearby trenches was 250 μm, and the width and depth of each trench were approximately 55 and 28 μm, respectively. The use of a LPSW was to secure reproducible LIBS measurement by facile spread of bile juice on the surface *via* multiple capillary actions without serious localization.

Each raw bile juice sample (100 μL) was initially mixed with a 100 μL LiCl solution (concentration: 100 ppm). LiCl powder was purchased from Sigma-Aldrich (product number: L9650). Incorporation of LiCl allowed utilization of a Li peak at 670.8 nm as a reference peak for intensity normalization. Then, 12 μL of the 1 : 1 mixture solution was dropped onto the LPSW and dried in an oven at 50 °C for 30 minutes before LIBS measurement. A commercial LIBS instrument (RT100-EC, Applied Spectra, Inc.) equipped with a flash-lamp-pumped Q-switched Nd:YAG laser and a five-channel spectrometer with a charge coupled device (CCD) was used to acquire LIBS spectra. The wavelength, pulse duration, and repetition rate of the laser were 1064 nm, 7 ns, and 10 Hz, respectively. Laser pulses with 18 mJ per pulse were focused on the sample surface to generate plasmas. The spot size of the focused laser beam producing plasmas was set to 150 μm in diameter by using a beam expander installed in the LIBS instrument. Emitted light from the laser-induced plasmas was collected and sent to the CCD spectrometer. The CCD detection gate with 1.05 ms width was delayed from the laser pulse firing by 0.5 μs. A 7 mm line scan was conducted for each LIBS spectrum by translating the sample stage at a rate of 1 mm s^{-1}. A total of 30 spectra were acquired for each sample.

Results and discussion

Evaluation of reproducibility of LIBS measurement using the LPSW

Since LiCl was chosen as an internal standard, it was necessary to examine whether LIBS peaks of LiCl interfere with LIBS peaks

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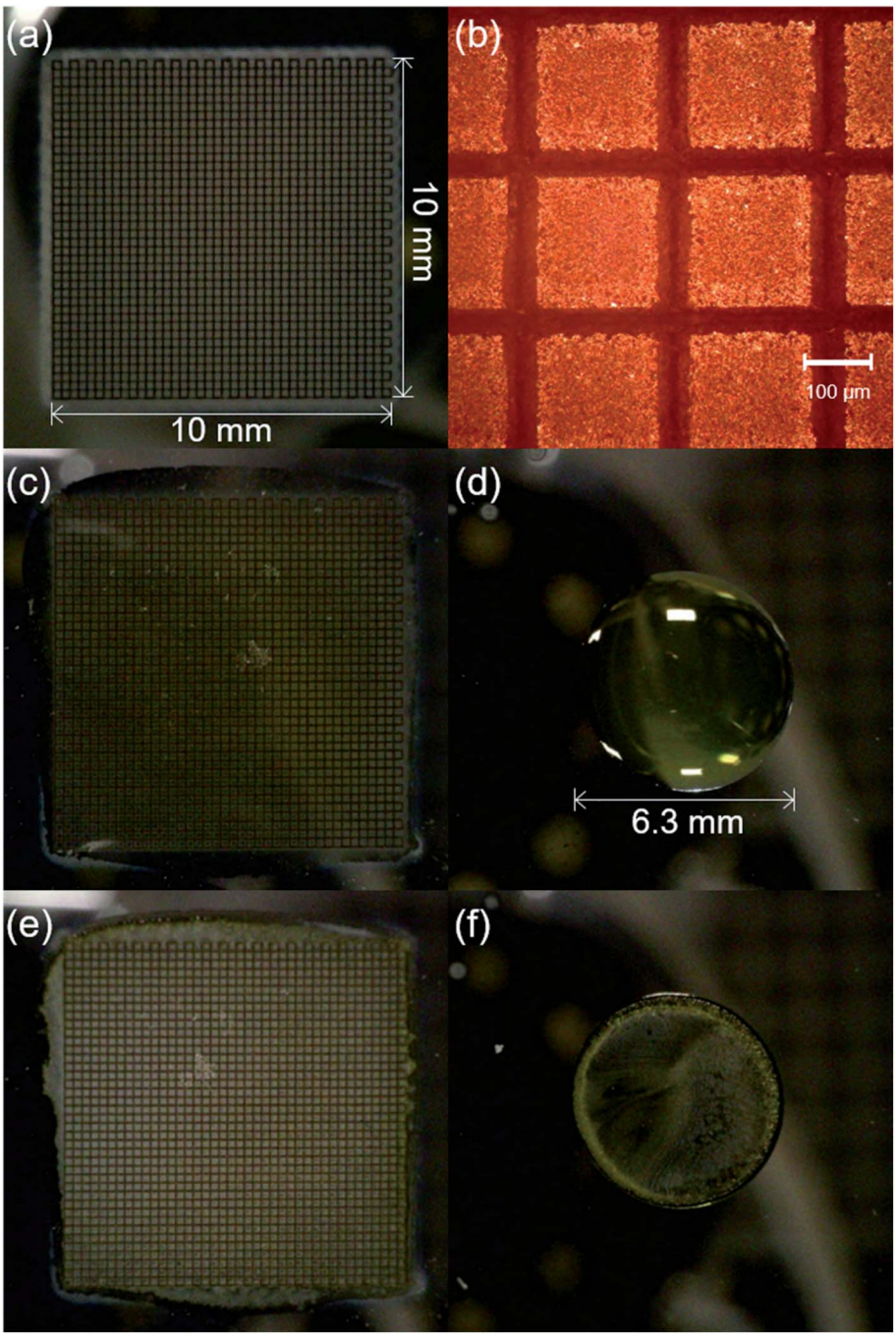

Fig. 1 Entire (a) and magnified (b) images of trenches on a LPSW, and pictures of a mixture of bile and LiCl dropped onto the patterned area of the LPSW (c) and bare Si surface (d). Pictures of the same samples dried on the patterned area (e) and bare Si surface (f) are also shown.

of raw bile juice or not. For this purpose, the LiCl (100 ppm) standard solution and bile of #1 GB polyp subject was separately dropped onto the patterned area of the LPSW, and their LIBS spectra were acquired by the 7 mm line scan mode as shown in Fig. 2(a) and (b), respectively. Each spectrum corresponds to the average of 30 spectra collected over the line scan. In the LiCl spectrum, five main Li peaks appear at 413.3, 460.3, 610.4, 670.8, and 812.6 nm, and the 670.8 nm peak is the most intense. Also, many Si peaks generated from the Si-based LPSW are present below 400 nm, and also additional small Si peaks appear at 634.7 and 637.1 nm. Since the Li and Si peaks appear, other major peaks of bile juice should not overlap with them for clear identification. Moreover, Na peaks at 589.0 and 589.6 nm as well as small K peak at 766.5 nm also appear. These peaks originated from the contained trace Na and K in the LiCl reagent, which are below 0.20 and 0.01%, respectively (denoted on the label of the reagent bottle, Sigma-Aldrich).

In the LIBS spectrum of the #1 GB polyp sample (Fig. 2(b)), Mg peaks at 279.6 and 280.3 nm, a Ca peak at 393.4 nm, Na peaks at 589.0 and 589.6 nm, and a K peak at 766.5 nm are

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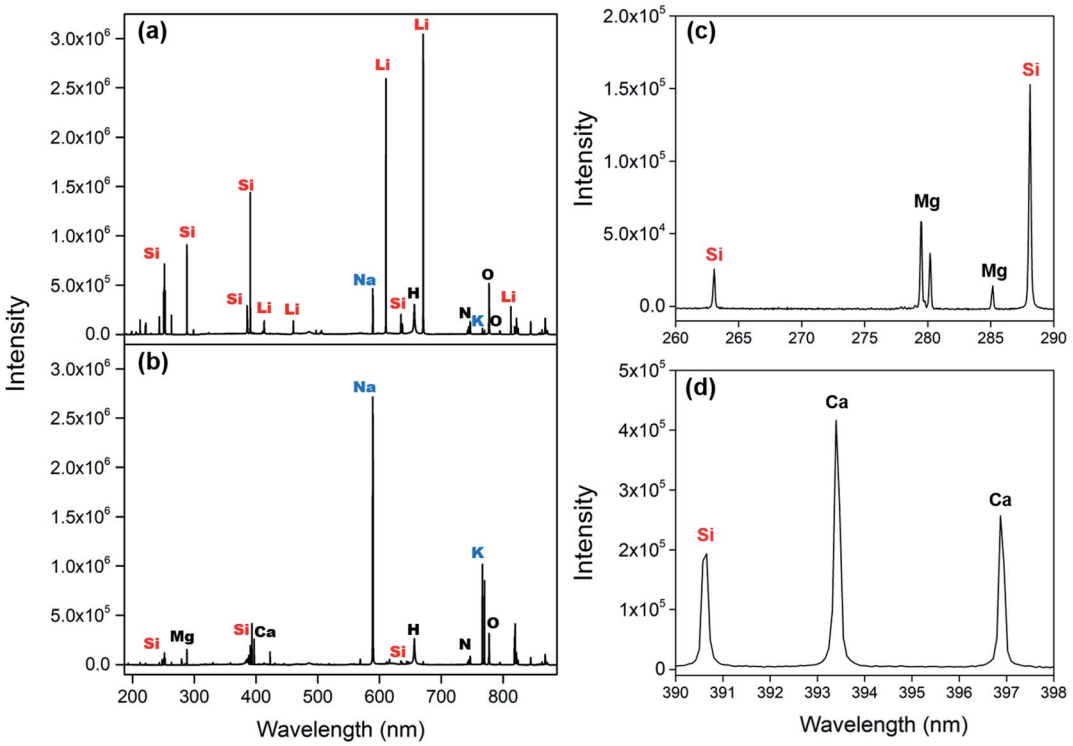

Fig. 2 LIBS spectra (180–887 nm range) of the LiCl (100 ppm) standard (a) and bile of #1 GB polyp subject (b) measured using the LPSW. The Mg (c) and Ca (d) peaks of the bile sample are highlighted for closer observation.

notable, and the Na peaks are the strongest. These peaks are typically present in diverse biological specimens as described in the introduction. Initially, the Mg, Ca, Na, and K peaks do not overlap the Li peaks, so the strongest Li peak at 670.8 nm can be adopted as a reference peak to normalize intensity fluctuation of the main element peaks. Next, the Na and K peaks apparently do not overlap the background Si peaks. Meanwhile, to closely examine whether the Mg and Ca peaks overlap with the Si peaks or not, the Mg peaks in the 260–290 nm range and the Ca peak in the 390–398 nm range are highlighted as shown in Fig. 2(c) and (d), respectively. In both cases, the nearby Si peaks are farther away from the Mg and Ca peaks without mutual overlapping. Overall, the Mg, Ca, Na, and K peaks of the bile sample are overlap-free of the Li as well as Si peaks.

Uniform spread of raw bile without serious localization on a substrate is an important factor in reproducible LIBS measurement. To observe the sample distribution on the LPSW, a 1 : 1 mixture (12 μL) of bile from #1 GB polyp subject and LiCl was dropped onto the patterned area, and subsequent sample dispersion on the substrate was examined, as shown in Fig. 1(c). The sample was quite evenly dispersed over the patterned area due to capillary forces generated by the crossed micro-channels. After drying, the sample distribution was maintained without drying-induced localization (Fig. 1(e)). In contrast, when the same sample was applied onto a bare Si wafer, formation of a hemispherical droplet was apparent (Fig. 1(d)), and the typical coffee-ring pattern was observed after drying (Fig. 1(f)). Reproducibility is substantially degraded when the coffee-ringed sample is measured for LIBS analysis. Therefore, the LPSW was effective in dispersing the bile sample over the substrate, which improved the reproducibility of bile measurement.

Fig. 3(a) and (b) show the average Li and Na peaks of the LiCl-incorporated #1 GB polyp (simply LiCl-bile) sample (blue spectra). The 30 spectra collected along the 7 mm line scan were averaged for examination. The shading on the average spectrum indicates the maximal magnitude of intensity variation among the 30 spectra. For comparison, the LIBS spectra of the LiCl standard (red) and the same bile sample without LiCl (black) are also displayed. When the Li peaks are analyzed (Fig. 3(a)), the intensity of the LiCl-bile sample is considerably lower than that of the LiCl standard due to the suppression of intensity by the complex and viscous organic matrix of bile. The intensities among the 30 spectra somewhat vary as noted by the shading. Meanwhile, no significant Li peak is observed in the spectrum of the bile sample with no LiCl standard.

In the case of the Na peak (Fig. 3(b)), the LiCl standard produces small peaks due to the contained trace Na as pointed out earlier. The presence of Na peaks from the internal standard is undesirable; however, the intensities are much weaker than those of the bile sample. Therefore, the Na peak intensities of the LiCl-bile sample are slightly higher than those of the bile sample without LiCl. Meanwhile, the magnitude of intensity variation (shading) among the 30 collected spectra along the line scan is considerably greater than the increased intensity by the addition of LiCl. The intensity of the Na peak induced from the LiCl standard will be consistent as long as the same standard is used for bile measurement, so its influence on subsequent analysis is expected to be insignificant. More importantly,

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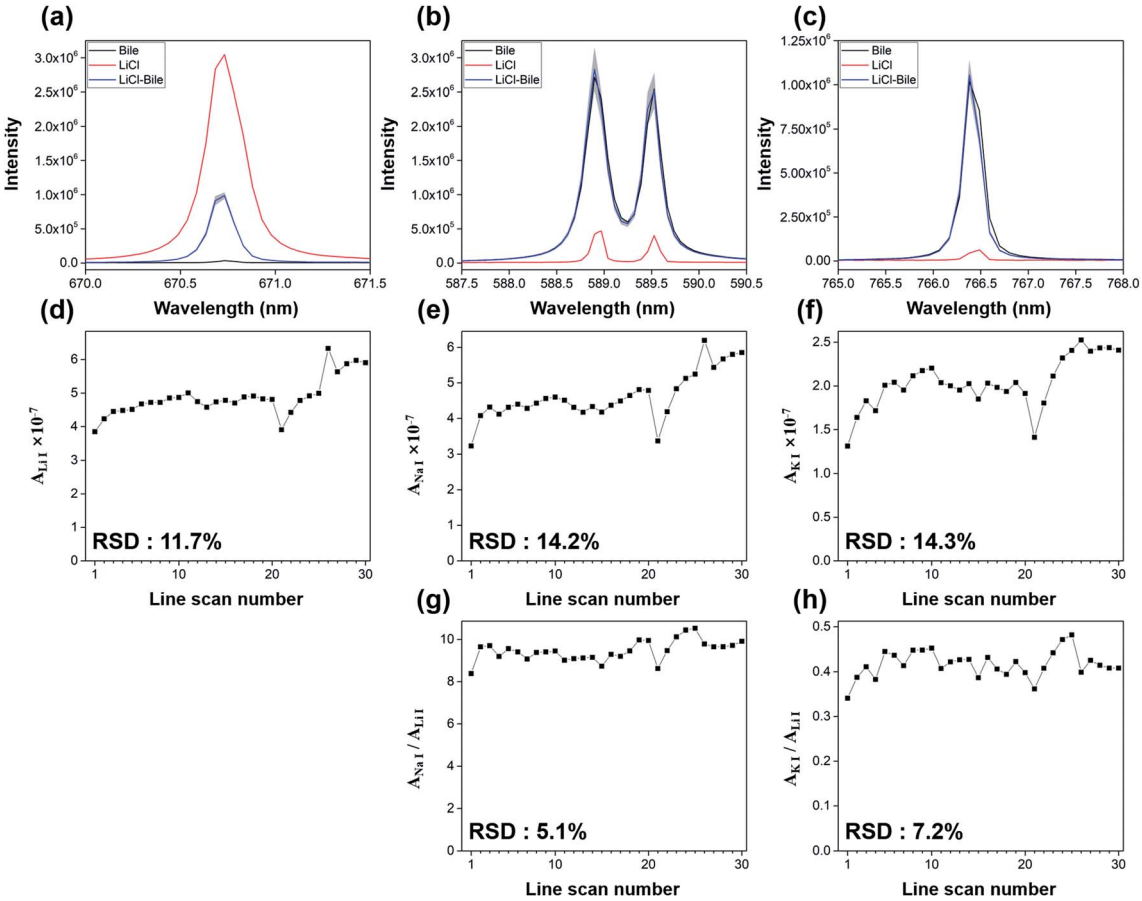

Fig. 3 Average Li (a), Na (b), and K (c) peaks in the spectra of the LiCl standard (red), LiCl-bile sample (blue), and bile sample with no LiCl standard (black). The shading on the average spectrum indicates the maximal magnitude of intensity variation among the 30 spectra collected along the 7 mm line scan. Variations of A_{Li} (d), A_{Na} (e), A_K (f), A_{Na}/A_{Li} (g), and A_K/A_{Li} (h) values in the 30 line-scanned spectra are shown for evaluation of reproducibility. The RSD in each case is designated in the corresponding plot.

the reduction of intensity variation among the 30 spectra becomes a crucial issue for reliable LIBS-based bile analysis.

Fig. 3(c) presents the K peaks at 766.5 nm in the spectra of the LiCl standard, LiCl-bile sample, and bile sample with no LiCl standard. In the spectrum of the LiCl standard, the K peak is very small, since the content of K in the LiCl powder is minute (below 0.01%). Therefore, the peak intensity of the LiCl-bile sample is nearly similar to that of the bile sample with no LiCl. This observation supports that the K peak originated from the standard would not seriously interfere with the K peak of the bile sample. By the way, Mg and Ca peaks were not observed in the spectrum of LiCl (corresponding data are not shown) since the LiCl powder did not contain these elements.

To quantitatively evaluate the reproducibility of the above LPSW-based LIBS measurement, the Na peak at 589.0 nm, K peak at 766.5 nm, and Li peak at 670.8 nm were selected. The areas of the Li, Na, and K peaks (A_{Li}, A_{Na}, and A_K) in the 670.0–671.5, 588.0–589.9, and 765.9–766.8 nm ranges, respectively, were calculated. Fig. 3(d)–(f) show the respective variations of A_{Li}, A_{Na}, and A_K values in the 30 line-scanned spectra. In each case, the peak areas are consistent over the 30 line scans, while largely fluctuating at the 21st and 26th positions. The calculated relative standard deviations (RSDs) were 11.7, 14.2, and 14.3% for A_{Li}, A_{Na}, and A_K, respectively, and are moderate. The observed variations originated from the non-perfect homogeneous spread of the sample on the LPSW as well as plasma temperature fluctuation. To take advantage of the Li internal standard, ratios of Na and Li peak areas (A_{Na}/A_{Li}) and K and Li peak areas (A_K/A_{Li}) were examined as shown in Fig. 3(g) and (h), respectively. The magnitudes of variation decreased for both ratios, and subsequent RSDs of A_{Na}/A_{Li} and A_K/A_{Li} values decreased to 5.1% and 7.2%, respectively. The compensation of intensity fluctuation by the Li internal standard peak helped to further enhance the reproducibility. The achieved RSDs were acceptable for LIBS analysis in general.

LIBS spectral features of bile juice samples according to GB disease

For each sample, the 30 replicate spectra acquired along the 7 mm line-scan were averaged, and the average spectrum was normalized to an intensity of 1.0 for the Li peak at 670.8 nm. In this manner, 30 spectra corresponding to the 6 normal, 17 GB polyp, and 7 GBC samples were produced, and the spectra within each sample group were separately averaged for

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examination. Fig. 4(a) shows the average spectra of the normal (black), GB polyp (red), and GBC (blue) groups. For closer observation, Fig. 4(b)–(e) show magnification of the Mg peaks at 279.6 and 280.3 nm, the Ca peak at 393.4 nm, Na peaks at 589.0 and 589.6 nm, and the K peak at 766.5 nm, respectively. The shading on each average spectrum indicates the maximal magnitude of intensity variation among the included samples. In all 4 cases, the peak intensities of normal and GB polyp samples were similar and greater than those of the GBC samples, in which the average peak intensities of these four elements were all lower. Nonetheless, thorough differentiation of GBC against normal/GB polyps using the peak intensities would be unrealizable since the shadings of the GBC spectra overlap those of the other groups.

For quantitative evaluation, the areas of the Mg, Ca, Na, and K peaks (A_{Mg}, A_{Ca}, A_{Na}, and A_K) in the 278.8–280.4, 393.0–393.8, 588.0–589.9, and 765.9–766.8 nm ranges, respectively, calculated from all the sample spectra were analyzed, as shown in

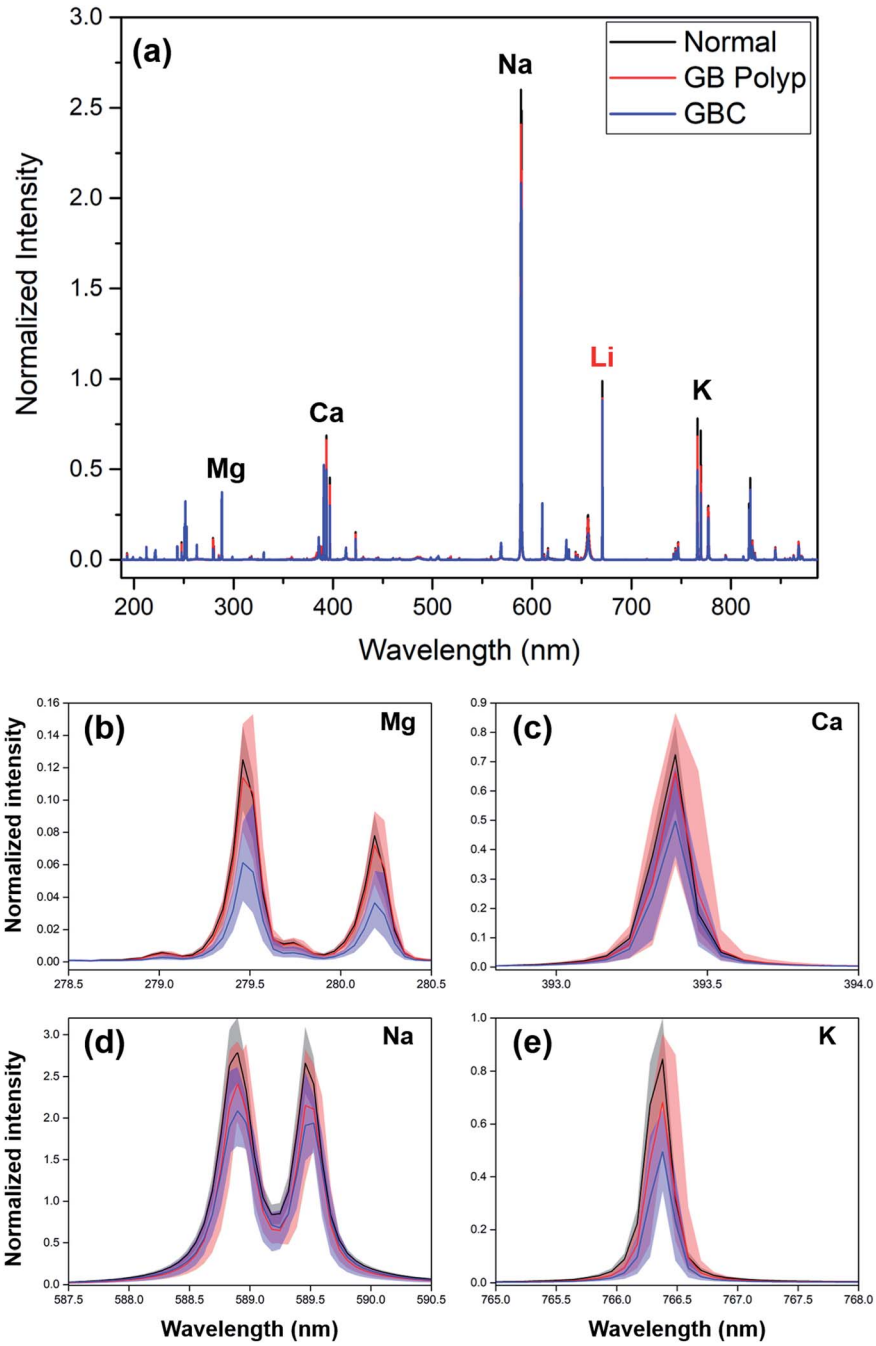

Fig. 4 Average LIBS spectra (187–887 nm range) of normal (black), GB polyp (red), and GBC (blue) groups (a) and the highlighted Mg peak at 279.6 and 280.3 nm (b), the Ca peak at 393.4 nm (c), Na peaks at 589.0 and 589.6 nm (d), and the K peak at 766.5 nm (e). The shading on each average spectrum indicates the maximal magnitude of intensity variation among the involved samples.

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Fig. 5. For each sample group, the corresponding average and standard deviation are displayed at the right of the peak area values. The black, red, and blue squares indicate normal, GB polyp, and GBC samples, respectively. As expected, all the A_{Mg}, A_{Ca}, A_{Na}, and A_K values of the GBC samples are generally lower than those of the normal/GB polyp samples. The statistical significance of the peak area difference among the groups was evaluated using the Wilcoxon test.[19] For the test, the normal and GB polyp groups were merged since the values from these two groups were not significantly different. The two groups of normal/GB polyps and GBC were subjected to statistical analysis. When the A_{Mg}, A_{Ca}, A_{Na}, and A_K values were used for the Wilcoxon test, the resulting *p*-values were 0.0002, 0.0124, 0.3268, and 0.0005, respectively (also shown in Fig. 5). At the 95.0% confidence level, the difference of A_{Na} values between the two groups was not significant (a *p*-value greater than 0.05), while the difference of A_{Ca} values was significant (a *p*-value smaller than 0.05). The lower A_{Mg} and A_K values in GBC were statistically significant with a confidence level greater than 99.9% (*p*-values smaller than 0.001).

The observed lower element concentrations in the GBC samples require explanation. Prior studies reported that the levels of total and individual bile acids in bile from GBC patients were lower compared to those in GB stone patients.[20,21] As reported, the GB stores and concentrates bile through absorption of water in the epithelial mucosa.[22] The expression of Cox-2 (cyclooxygenase-2) in the GB epithelium was elevated with the occurrence of cancer and carcinogenesis;[23,24] the increase of Cox-2 thickened the GB wall and degraded water absorption by the mucosa.[25] The lower levels of the metals in the case of GBC can be attributed to dilution of bile by the reduced water absorption. Meanwhile, since the A_{Mg} and A_K values are further lower in GBC, it appears that, along with bile dilution, other factors are involved.

A ratio of two peak areas is alternatively a valuable index for GB disease discrimination since multiple elements simultaneously play important roles in various metabolic pathways, and their quantitative balance would often be disease dependent. Therefore, peak areas from two of the four elements were chosen in a combinatorial fashion, and subsequent peak area ratios were evaluated for discrimination. Fig. 6 shows A_{Mg}/A_{Ca}, A_{Mg}/A_{Na}, A_{Mg}/A_K, A_{Ca}/A_{Na}, A_{Ca}/A_K, and A_{Na}/A_K values corresponding to all the samples. Again, the corresponding average and standard deviation are presented at the right of each peak area ratio data, and relevant *p*-values are indicated in the figure. At the 95.0% confidence level, the difference of A_{Ca}/A_K values between the two groups is not significant (*p*-value: 0.1285, much greater than 0.05), while the difference of A_{Ca}/A_{Na} values is significant (*p*-value: 0.0108, smaller than 0.05). The differences in A_{Mg}/A_{Ca} and A_{Mg}/A_K values are significant at confidence levels greater than 99.0% (*p*-values smaller than 0.01). Moreover, with a higher confidence level of 99.9% (*p*-values smaller than 0.001), the differences in A_{Mg}/A_{Na} and A_{Na}/A_K values are significant, and A_{Na}/A_K values that provide the smallest *p*-value of 0.0001 are the most discriminant of the normal/GB polyp and GBC groups.

The origin of increased A_{Na}/A_K values in the GBC samples needs to be addressed. The most plausible cause would be a claim by Pantellini *et al.* Salification of K^+ in DNA and RNA is a prerequisite for keeping their structures stable in a biological environment. Meanwhile, the replacement of intracellular K^+ with Na^+ (increase of the Na/K ratio) stimulates the opening of the valence bonds of the bases, thereby altering the corresponding genetic information and subsequently producing anomalous polymerized structures (equivalent to neoplasms) in the cell cytoplasm.[26] Also, Jansson *et al.* reported the existence of a positive relationship between the cancer rate and Na/K ratio, and the relationship was more apparent for cancers of digestive organs.[27] Overall, intracellular K and Na concentrations are probably related to development of cancer, and the

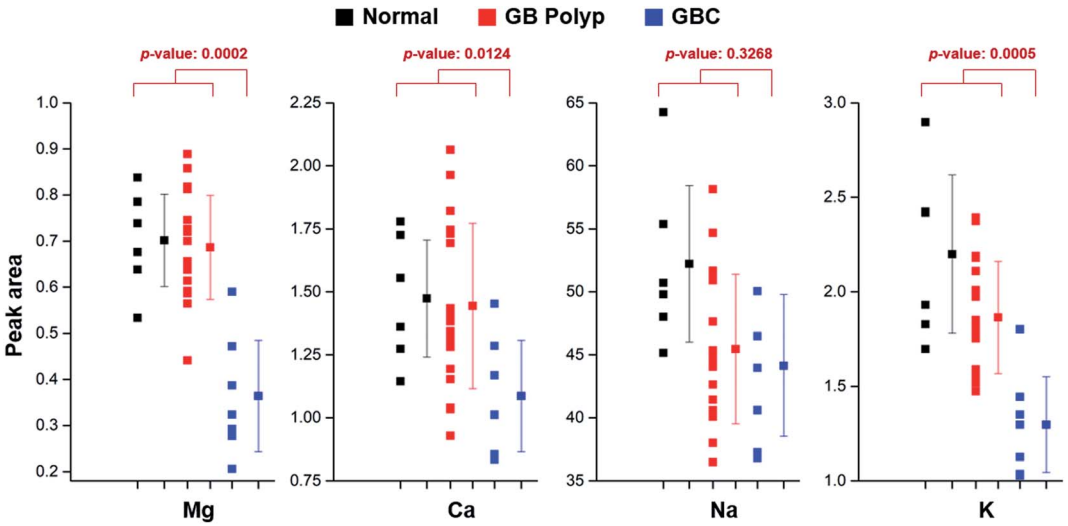

Fig. 5 The A_{Mg}, A_{Ca}, A_{Na}, and A_K values obtained from the LIBS spectra of all samples. The corresponding averages and standard deviations are displayed at the right of each peak area data. The resulting *p*-value based on the Wilcoxon test in each case is designated in the corresponding plot.

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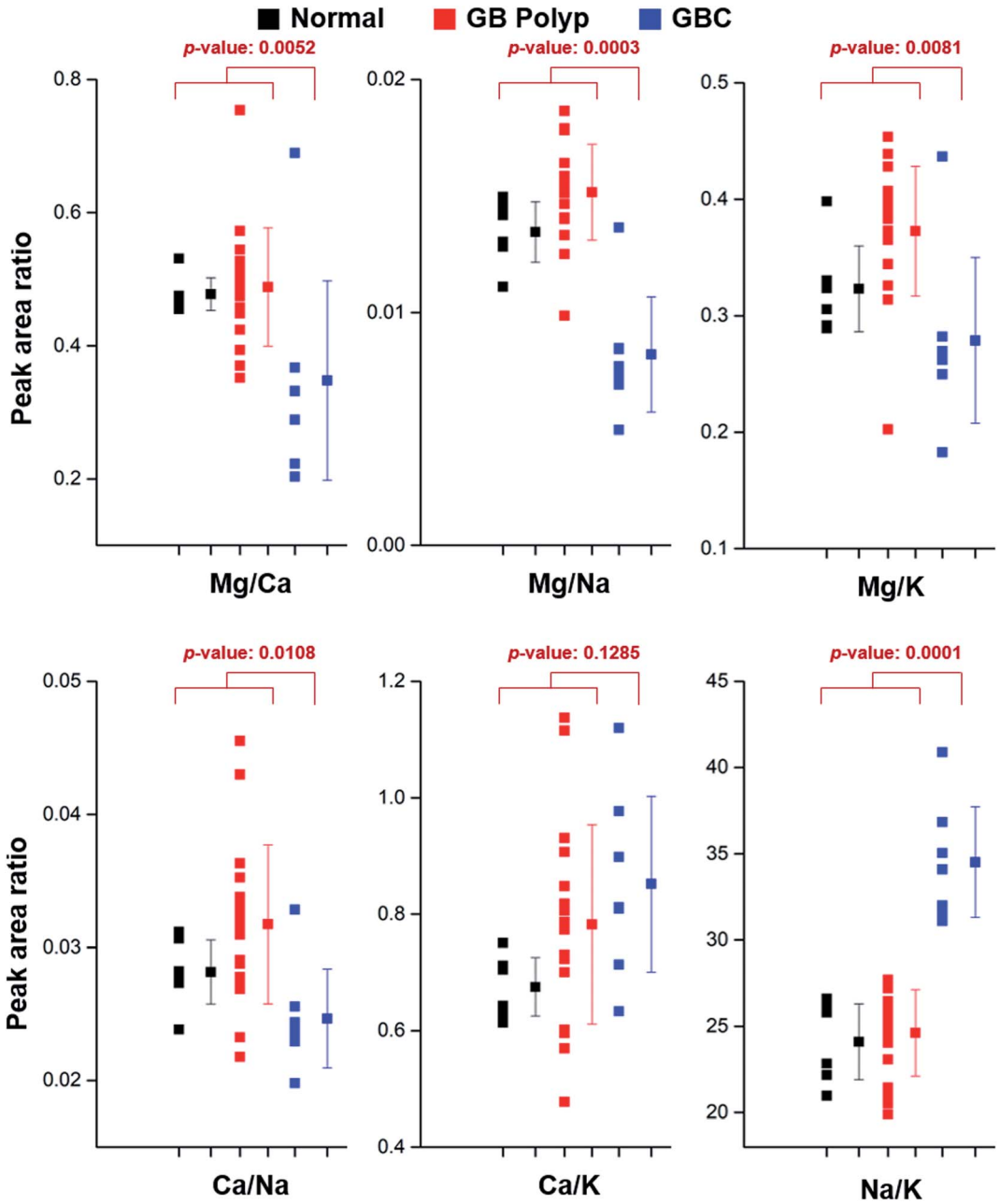

Fig. 6 The A_{Mg}/A_{Ca}, A_{Mg}/A_{Na}, A_{Mg}/A_{K}, A_{Ca}/A_{Na}, A_{Ca}/A_{K}, and A_{Na}/A_{K} values obtained from the LIBS spectra of all samples. The corresponding averages and standard deviations are displayed at the right of each peak area ratio data. The resulting p-value based on the Wilcoxon test in each case is designated in the corresponding plot.

increase of the Na/K ratio is a probable cause of elevated cancer occurrence, such as GBC in this study.

Conclusion

The elemental compositions of the bile juice samples differed according to the pathological condition of the GB, such as the ratio of Na and K peak areas, and the dissimilar composition was effectively recognized by simple LIBS measurement for identification of GBC against normal/GB polyps. The LPSW enabling facile spread of viscous bile over the surface without noticeable localization made the LIBS analysis viable for discrimination. Meanwhile, further investigations are necessary to drive the LIBS-based bile analysis scheme closer to real clinical analysis. The number of employed bile samples must increase to demonstrate clinical applicability of the method with statistical evaluation. More in-depth investigations on the pathological origins of the higher Na/K ratio, the lower Mg/Na ratio, and the further lower Mg content in bile juice of GBC subjects are necessary. In addition, the combination of elemental information with the molecular spectroscopic signature of bile juice would be a synergetic strategy to enhance discrimination of GB diseases. The future work noted above is underway in this research group.

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Conflicts of interest

There are no conflicts to declare.

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